

# Social Media Agent

Intelligent Multi-Channel Companion for Real-Time Social Intelligence

## 1. The Challenge

Brands, influencers, and organizations struggle to keep up with the fast-paced and fragmented world of social media. Content trends evolve hourly, audience sentiment shifts rapidly, and competitor strategies are difficult to track.

Managing multiple platforms often leads to scattered insights, missed engagement opportunities, and ineffective campaigns. The core challenge is not only to collect social media data but also to analyze, predict, and recommend actions in real time.

## 2. The Objective

Build an agentic AI application that acts as an intelligent social media companion. The solution should encompass four key capabilities:

### 1. Fuse Multichannel Data

Collect and synthesize content, comments, hashtags, and competitor posts from platforms like **X (Twitter)**, **LinkedIn**, **Instagram**, and **Facebook** into unified actionable insights.

### 2. Enable Multimodal Interaction

Accept diverse user inputs (text, images, or voice prompts) and generate creative post drafts, captions, or full campaign ideas closely aligned with brand tone.

### 3. Create a Predictive & Agentic Layer

Forecast trending hashtags, audience sentiment shifts, and competitor engagement patterns while autonomously suggesting optimal posting schedules.

### 4. Visualize Social Media Pulse

Provide a comprehensive dashboard featuring real-time analytics—engagement trends, influencer mapping, sentiment tracking, and campaign performance forecasts.

### 3. Technical Requirements & Architecture

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To successfully deliver an agentic social media application, the solution must meet the following mandatory tech stack specifications:

#### Mandatory Tech Stack

IBM Langflow / IBM Orchestrate

IBM Granite / Foundation Models

**Architectural Focus:** Leverage IBM Watsonx / Langflow workflows to orchestrate multimodal AI agents, perform automated data extraction, execute real-time sentiment scoring, and generate targeted multichannel media content.